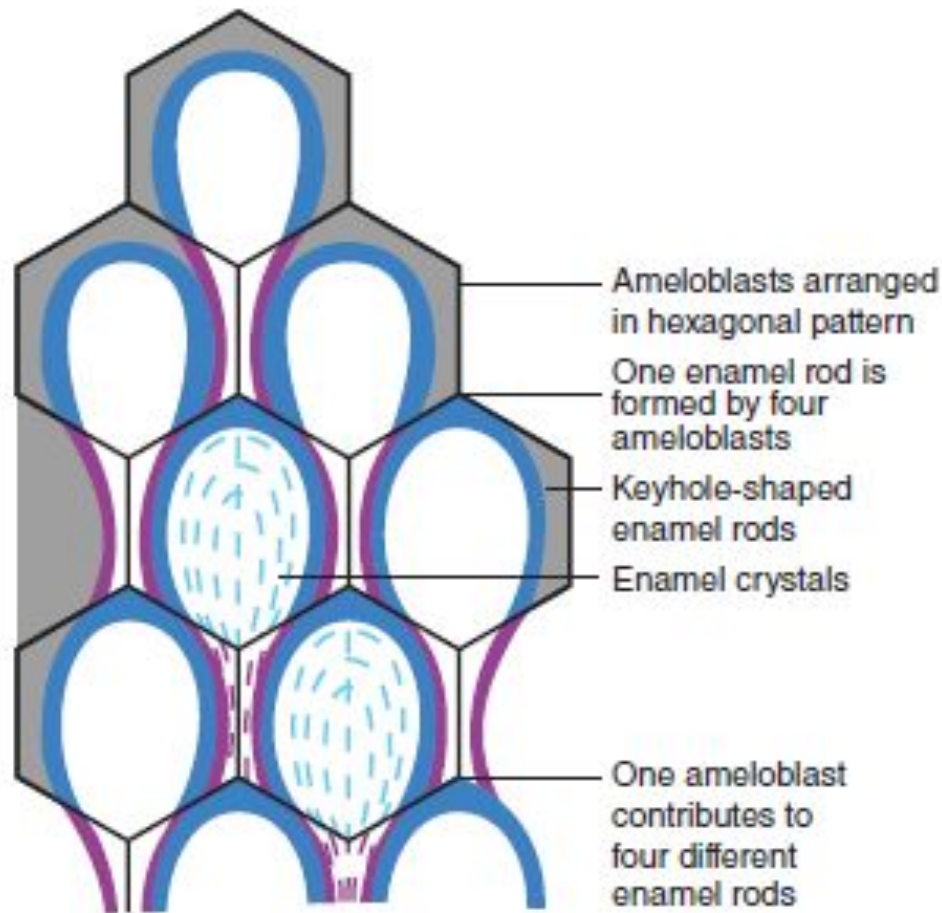




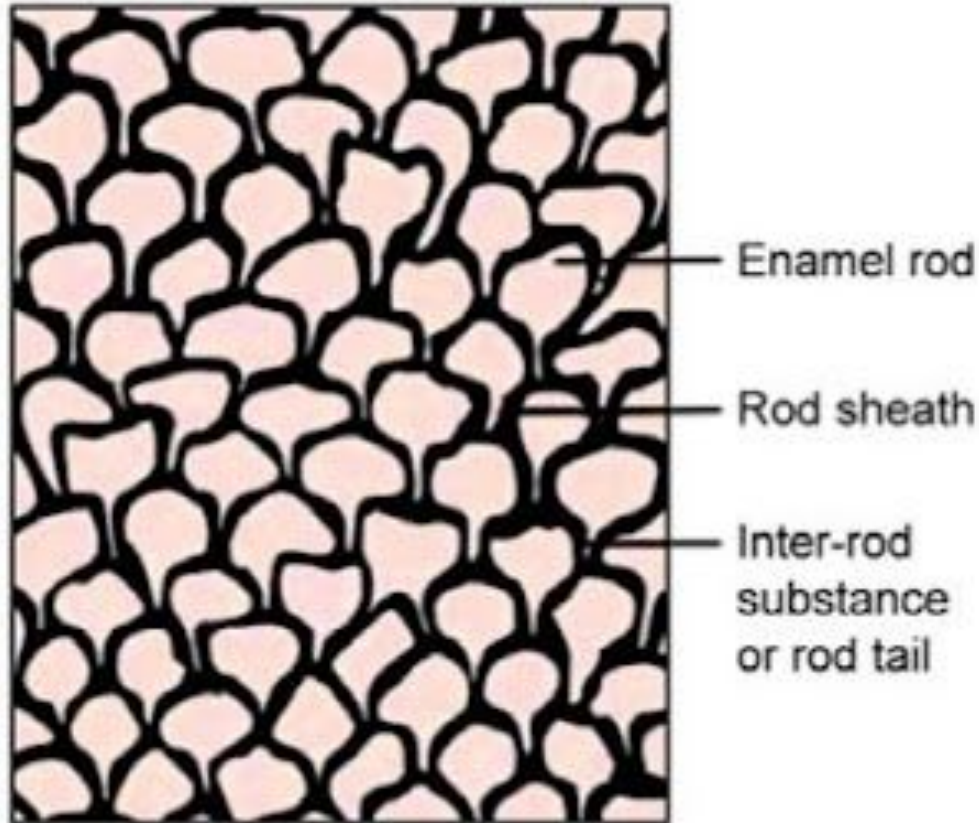
ENAMEL

Prof Farzeen Waseem

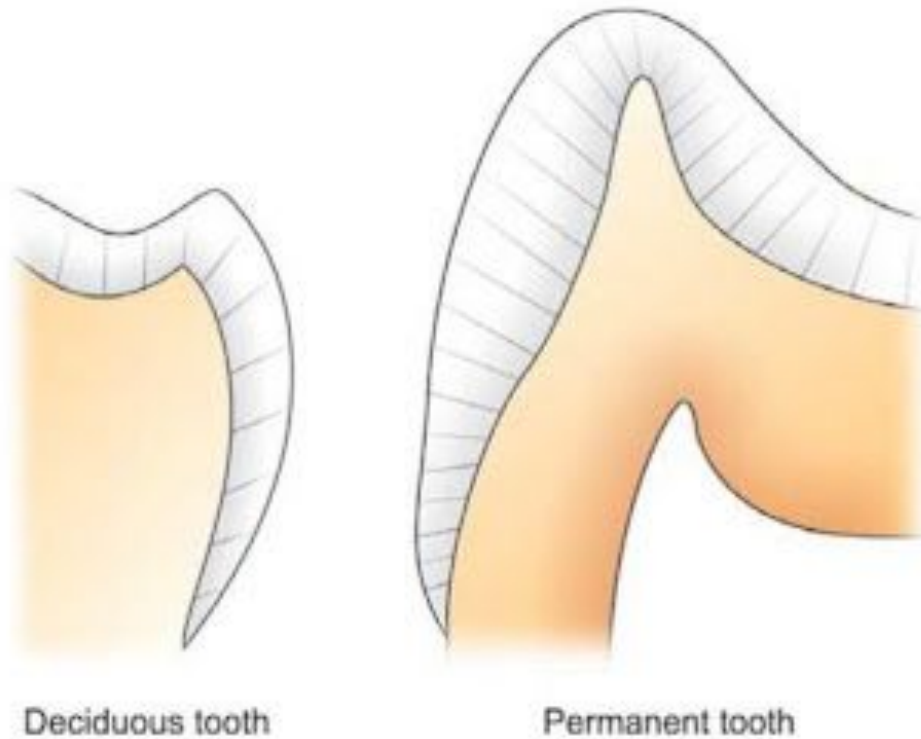
Oral BIOLOGY ,DIKIOHS-DUHS



An enamel rod is a long, thin structure extending from the dentino-enamel junction to the surface of enamel.



- Each rod is formed by four ameloblasts.
- One ameloblast forms the rod head, a part of two ameloblasts form the neck, and the tail is formed by a fourth ameloblast.
- Each ameloblast contributes to four different rods.
- In a cross-section of human enamel, many rods resemble fish-scales.



Direction of Rods

In general, the rods are directed at right angles to the dentino-enamel junction and the tooth surface.

In cervical regions of deciduous and permanent teeth, the directions of enamel rods are different.

In permanent teeth: At the central part of the fossa and pits, the rods turn oblique to nearly horizontal.

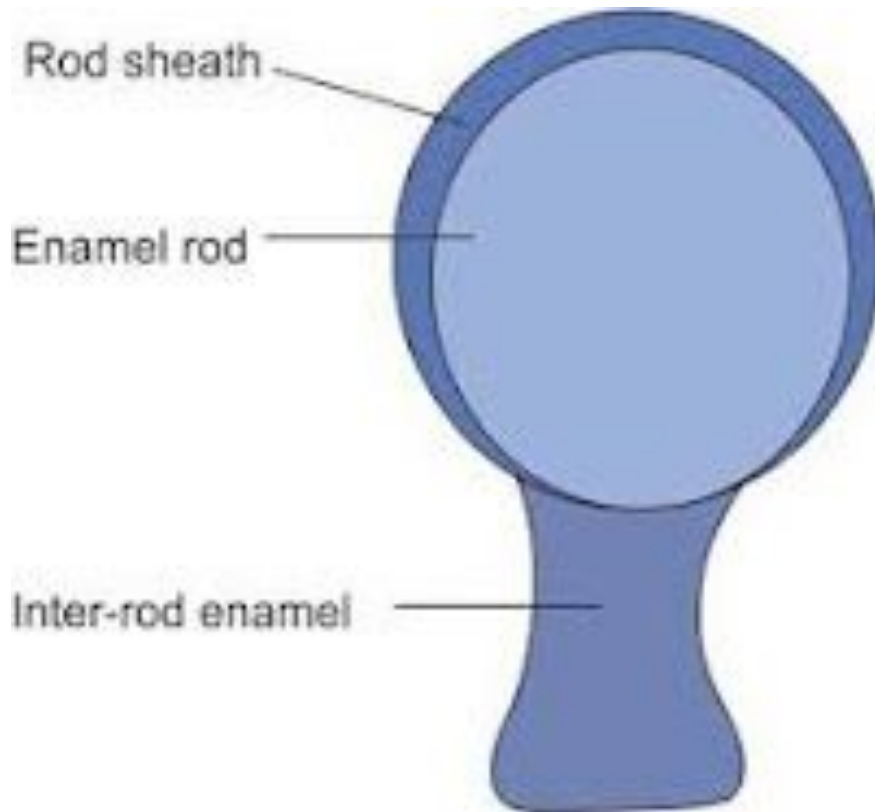
Near the tip of the cusp and incisal edge, the rods gradually change to increasingly oblique direction.

Gradually they become nearly vertical at the tip of the cusp and incisal region.

In the cervical region, the rods run slightly apically from dentin surface to outer enamel surface

- In deciduous teeth: The arrangement of rods in the occlusal two-thirds in deciduous teeth is similar to that in permanent teeth.
- At the cervical and central part of the fossa and pit, they are nearly horizontal.
- Submicroscopic structures: The enamel rods have an average diameter of 4 microns.
- The diameter of the rod increases from the dentino-enamel junction towards the outer surface of enamel in a ratio of about 1 : 2.
- The number of enamel rods has been estimated as ranging from 5 million in the lower lateral incisor to 12 million in the upper first molar.
- The rods are larger at cusp tips and shorter at the cervical region.
- The head or body of the enamel rod is the broadest part and is about 5 microns wide, and the elongated thinner portion called as tail is about 1 micron wide.
- The rod, including both head and tail, is about 9 microns long.
- In cross-section, the enamel rods appear as round, hexagonal or oval.
- Recent studies with electron microscope have shown that a more common pattern of enamel rods in cross section looks like a keyhole or paddle-shaped prism.

ROD SHEATH



- **distinct thin layer is seen peripheral to the rods.**
- **stains darker**
- **less calcified and contains more organic substance like enamel protein.**

This layer is known as rod sheath.

Hunter Schregar Bands

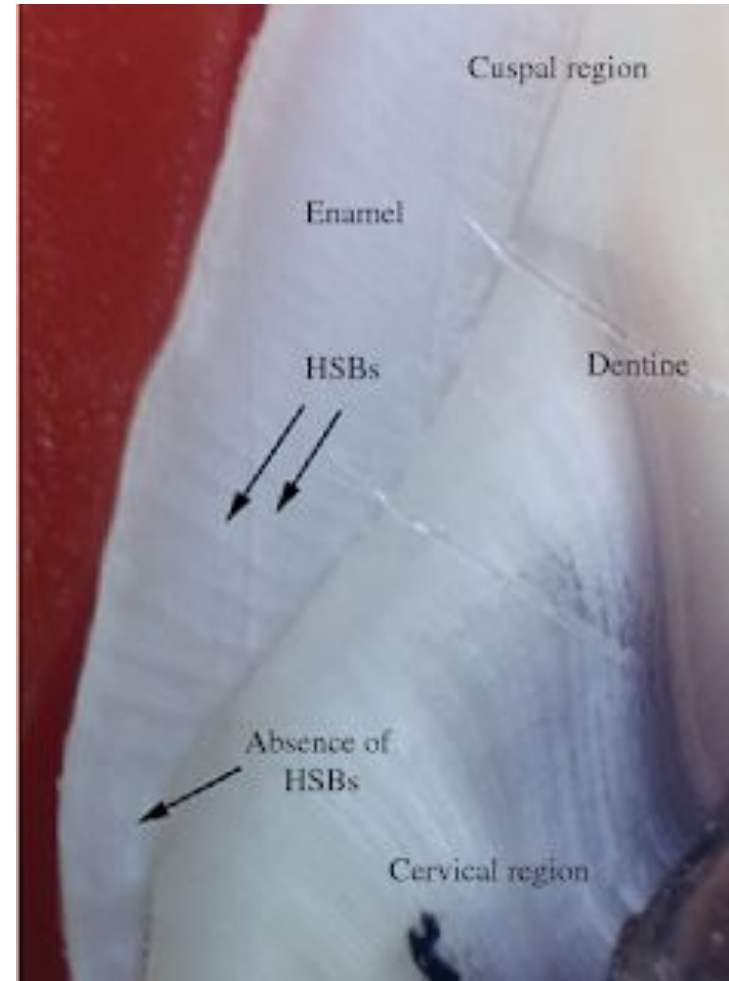
These are alternating dark and light bands which are best visualized in longitudinal ground section under oblique reflected light.

It is produced solely by changes in the rod direction.

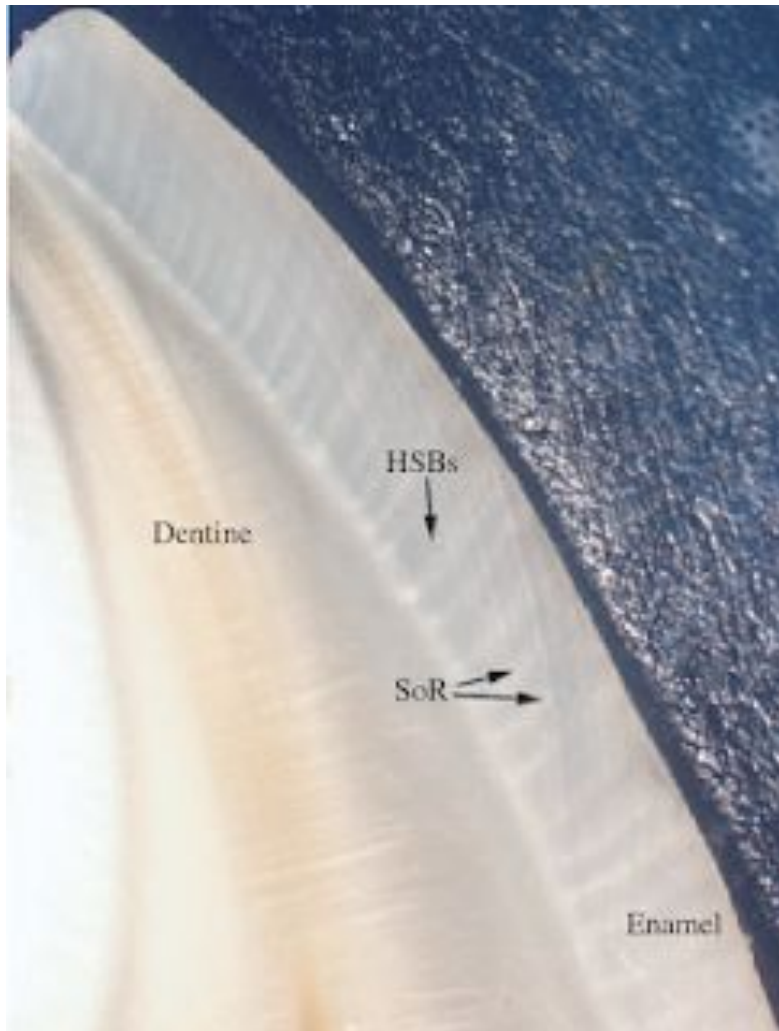
The light bands are referred to as diazones and the dark bands are called as parazones.

The angle between the diazones and parazones is approximately of 40 degrees

- 1. Change in Rod Direction**
- 2. Different content in Organic Material**
- 3. Difference in Permeability**



INCREMENTAL LINES



These are rhythmic successive apposition of layers of enamel during formation of the crown.

When a ground section of a tooth is seen under a light microscope, concentric brown lines are seen in the enamel. These are called incremental lines of Retzius or Striae of Retzius.

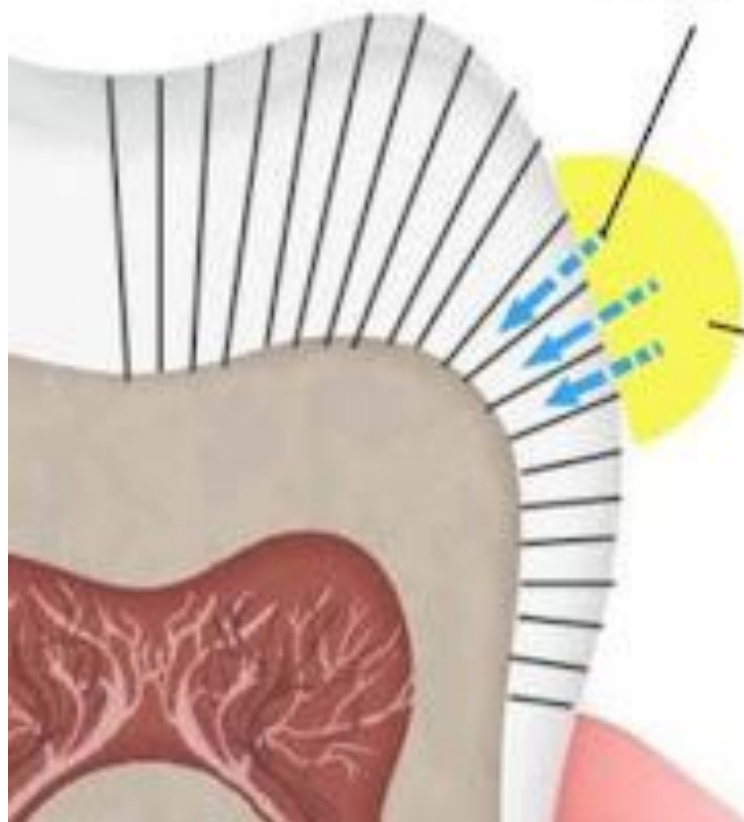
The incremental lines (striae) of Retzius are more frequently seen in permanent teeth and less frequently in deciduous teeth and prenatal enamel.

Prismless

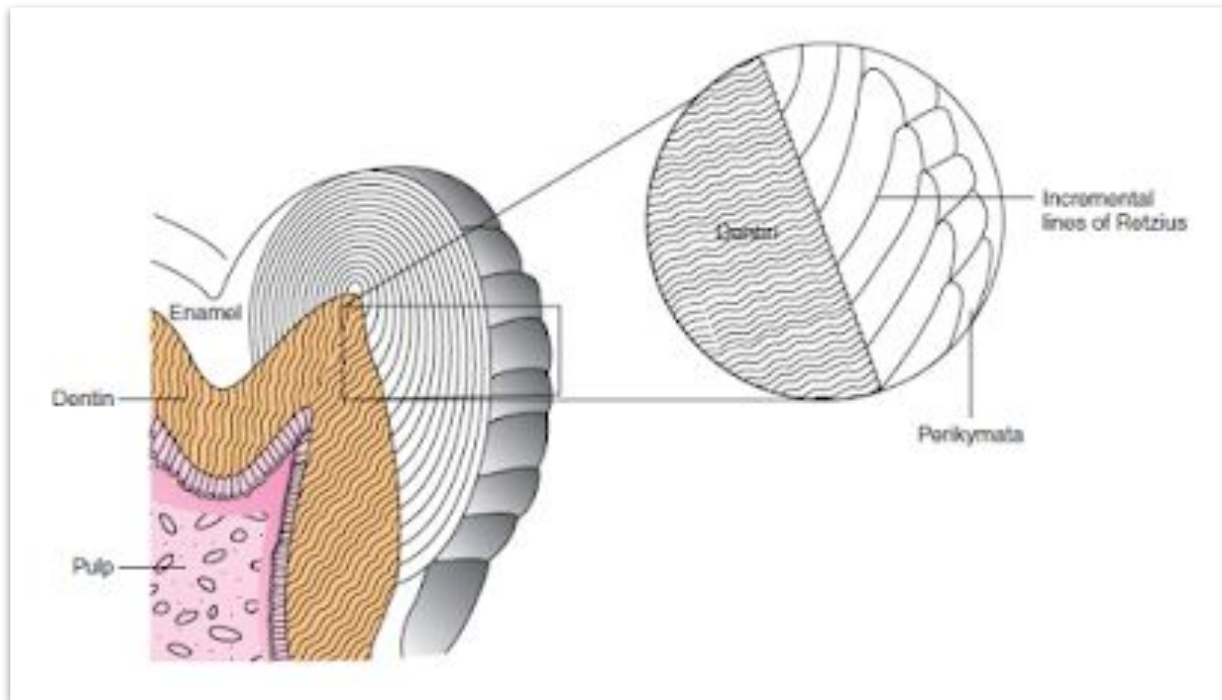
- Structureless Outer Enamel Layer
- 30 microns thick found most commonly towards the cervical area and less often on cusp tips.
- This structure less layer is called **PRISM LESS ENAMEL** and found in all deciduous teeth and in 70 percent of the permanent teeth.

blue arrows show bacteria
and their acids penetrating
enamel through small
channels in the enamel
layer

yellow area is a clump of
plaque or bacteria on the
tooth surface

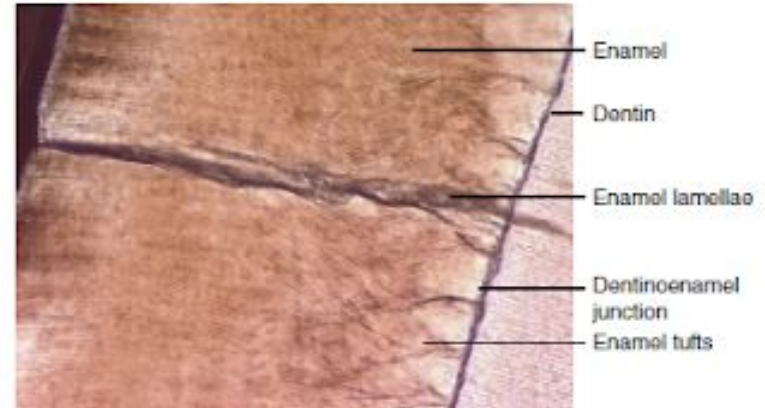


PERIKYMATA



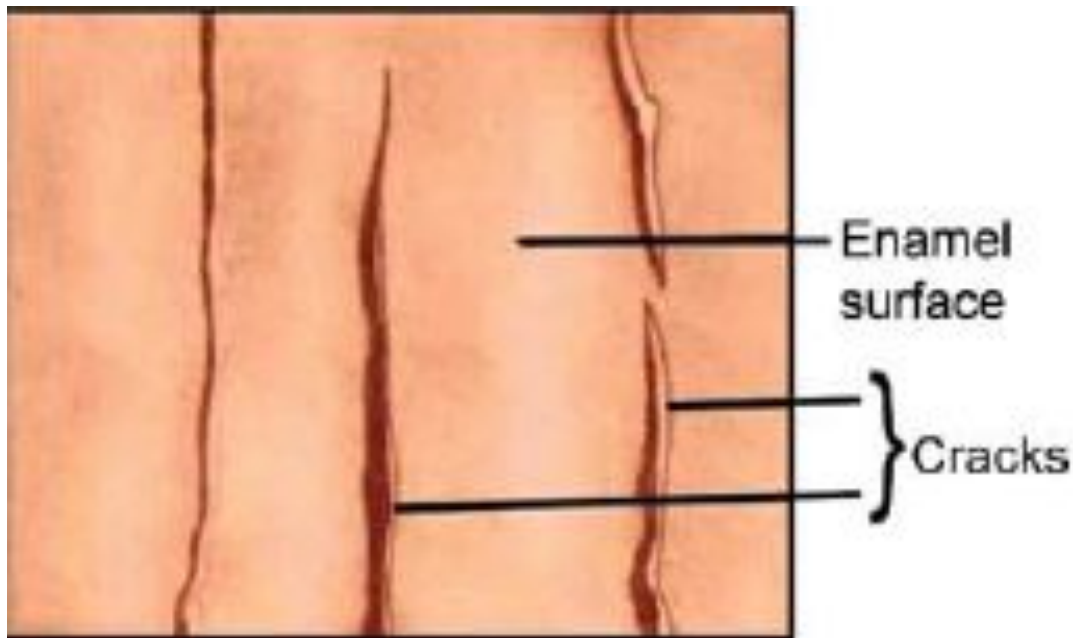
- Wave-like, transverse grooves, shallow furrows external manifestation of incremental lines of Retzius.
- Continuous around a tooth and usually lie parallel to each other and to the cemento-enamel junction.

ENAMEL LAMELLA



- Enamel lamellae are very thin, leaf-like structures, sometimes visible to naked eye.
- They extend from the enamel surface towards the dentino-enamel junction, rarely extending into dentin.
- The enamel lamellae contain mostly organic material.
- Enamel lamellae can be differentiated into three types:
- Type A lamellae composed of poorly calcified rod segment.
- Type B lamellae composed of degenerated cells.
- Type C lamellae arising in erupted teeth where the cracks are filled with organic matter and debris from saliva.
- Type A is restricted to enamel and type B and C may reach the dentin

ENAMEL CRACKS



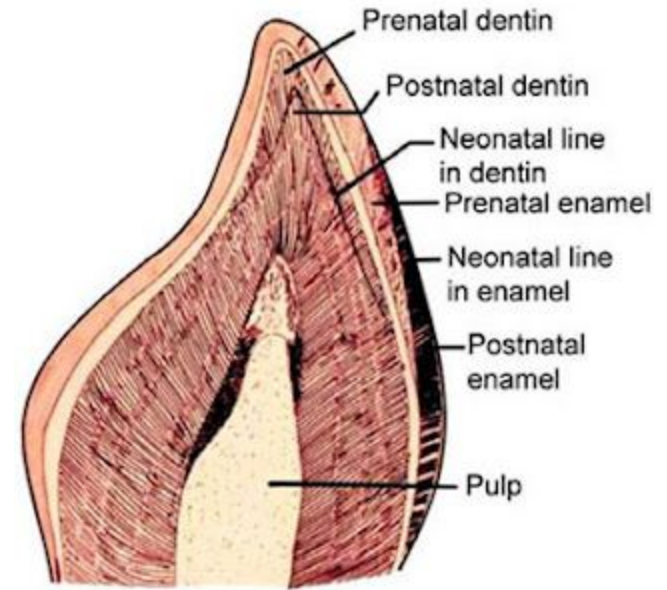
Narrow, fissure-like structures that are present on almost all surfaces.

They are actually the outer edges of enamel lamellae.

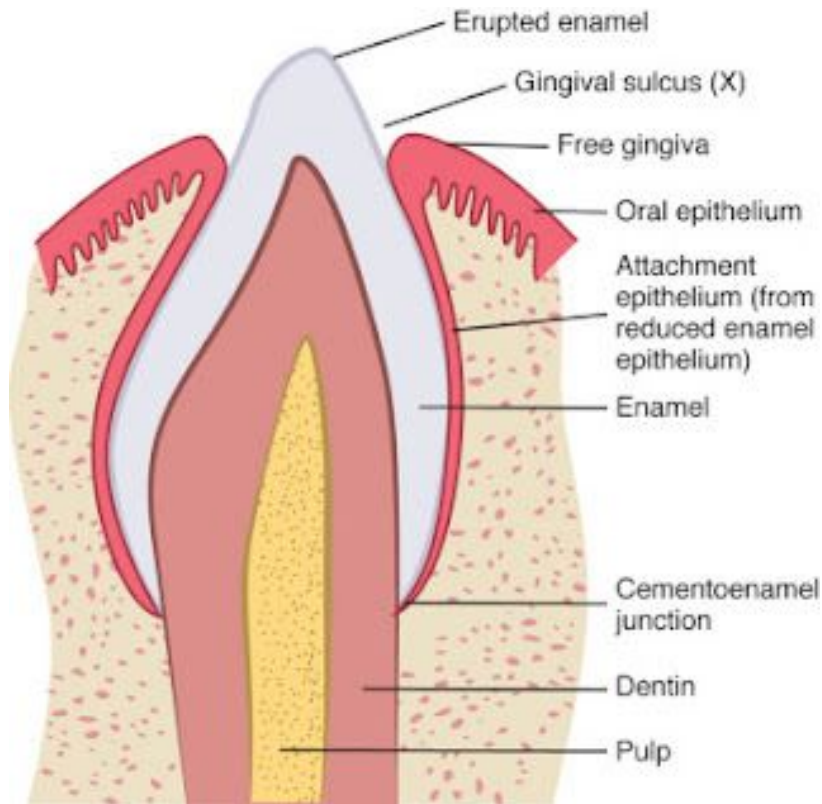
They originate from dentino-enamel junction and run at right angles to it.

NEONATAL LINES

- In deciduous teeth, the enamel develops partly before and partly after birth.
- The line or boundary between the two portions of enamel in deciduous teeth is known as neonatal line or neonatal ring.
- It appears due to the abrupt change in the environment and nutrition of the newborn (infants).
- It is an accentuated incremental line of Retzius.

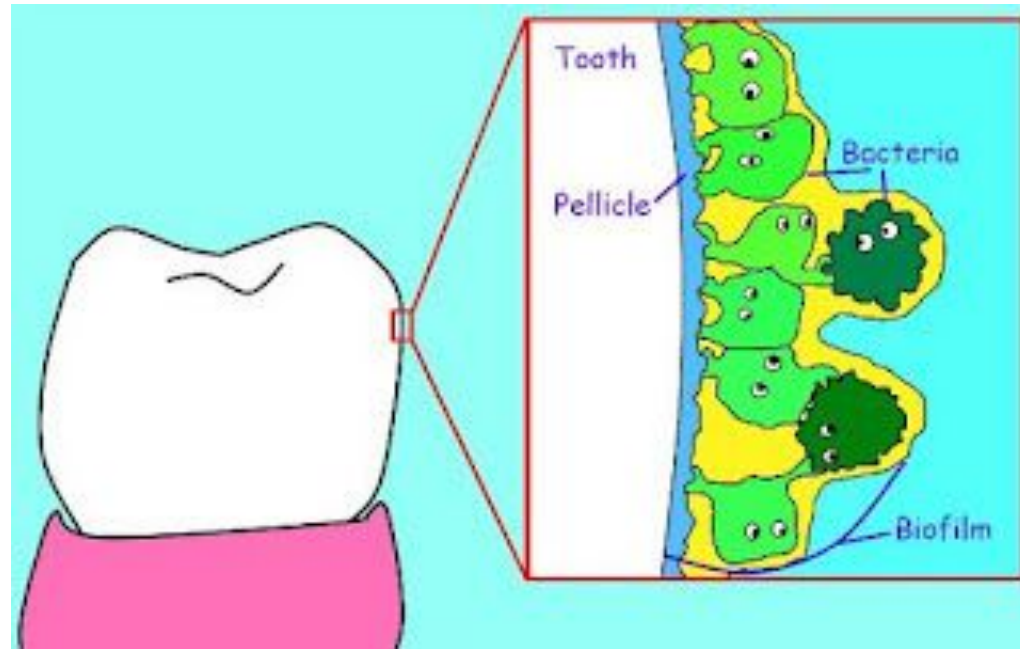


Nasmyth's Membrane (Primary Enamel Cuticle)



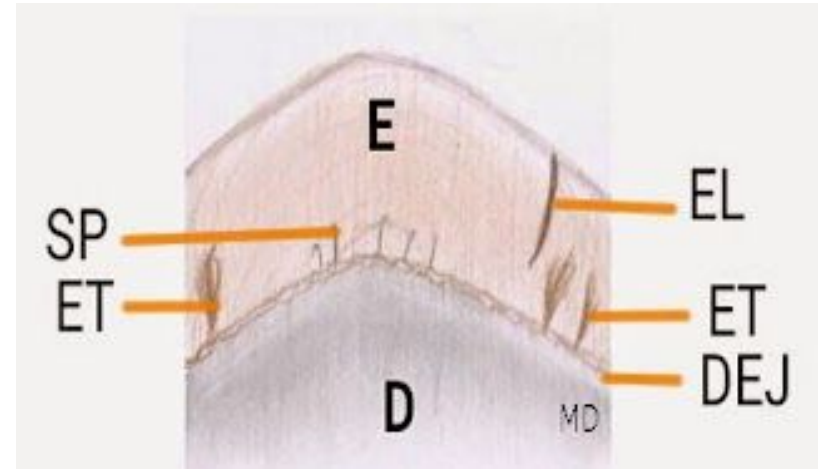
Nonmineralized usually found between the epithelium of dentogingival junction and the enamel surface. It is formed by an accumulation of basal lamina material produced by the junctional epithelium of the dentogingival junction. The final act of the ameloblast cell is secretion of a layer covering the end of the enamel rod.

SALIVARY PELLICLE



After tooth is cleaned, salivary proteins and glycoproteins having strong affinity for enamel get adsorbed to the enamel surface very quickly and form a very thin layer called the salivary pellicle.

ENAMEL TUFT



Enamel tufts are hypocalcified enamel rods and interprismatic substance that originates at the dentinoenamel junction and extends into enamel for about one-third to one-fifth of its total thickness.

They are known as enamel tufts because they look **like tufts of grass** projecting into enamel.

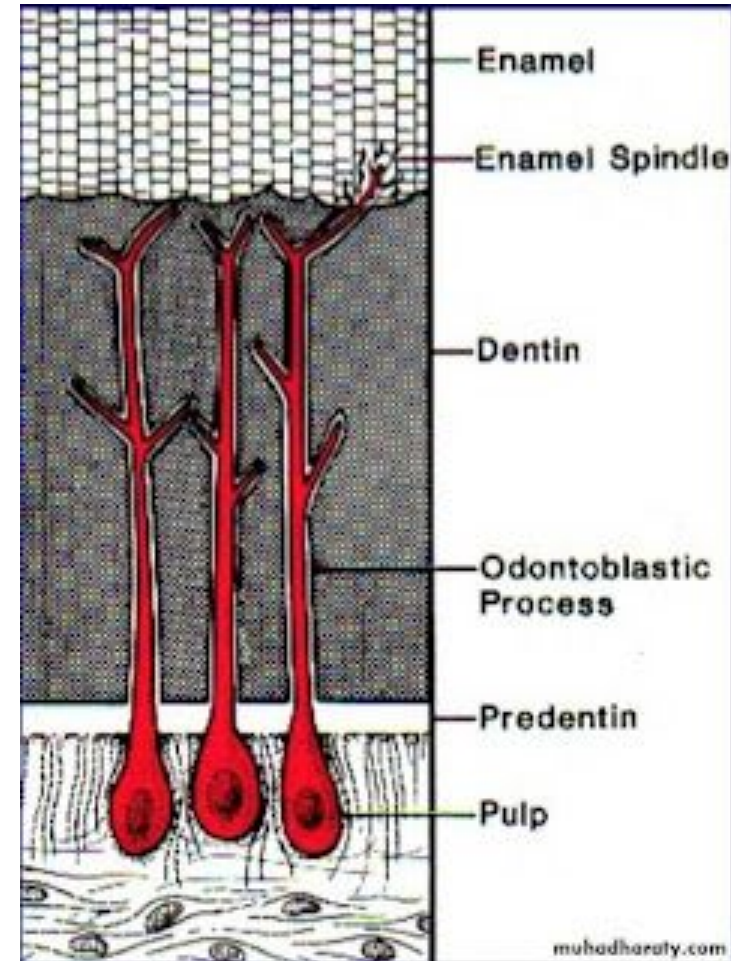
DENTINO ENAMEL JUNCTION

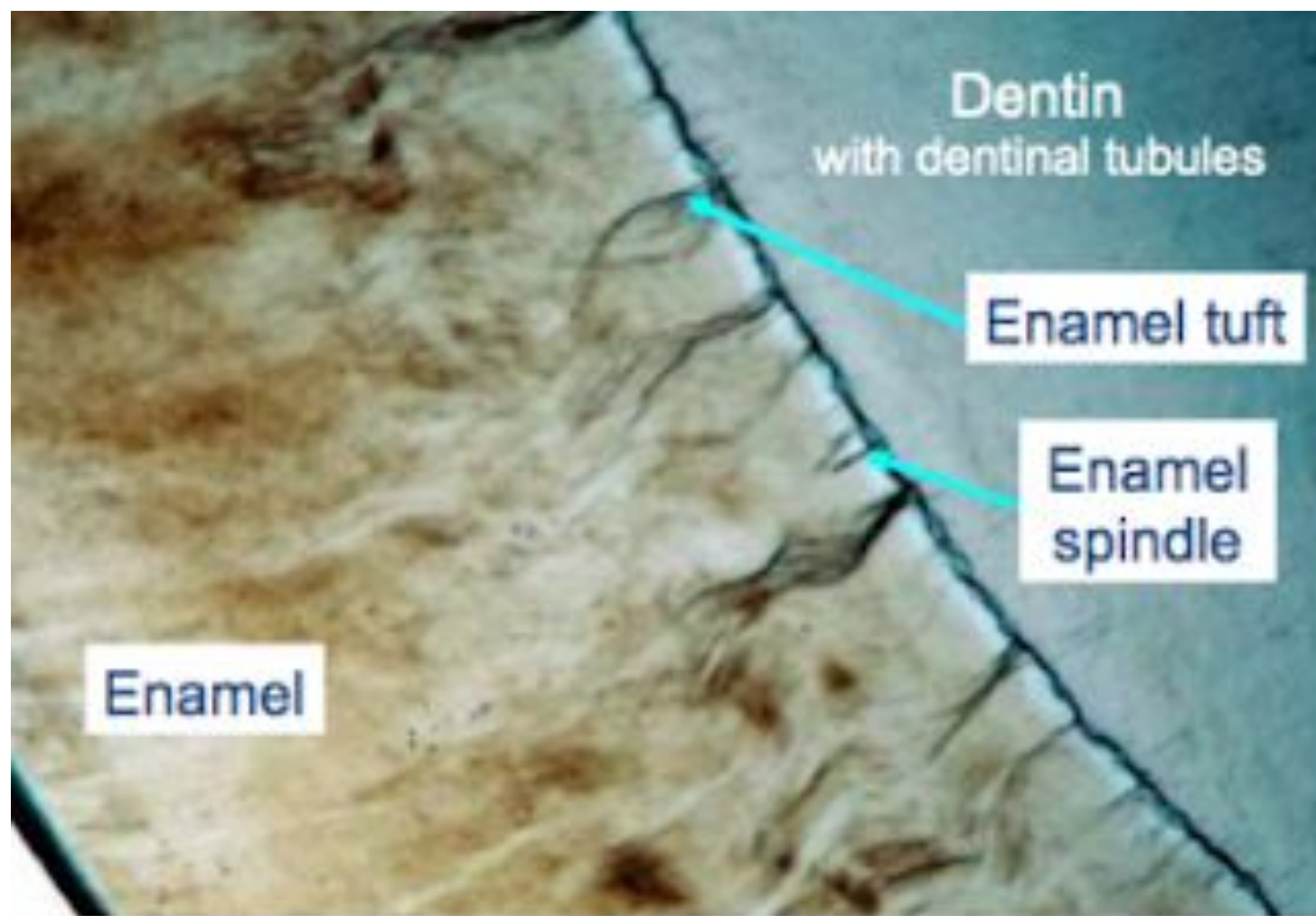


- The dentinoenamel junction is a scalloped interface between the enamel and dentin.
- Dentin has pitted surface, which supports the enamel.
- Small curved projections of enamel fit into small concavities of the dentin.

ENAMEL SPINDLE

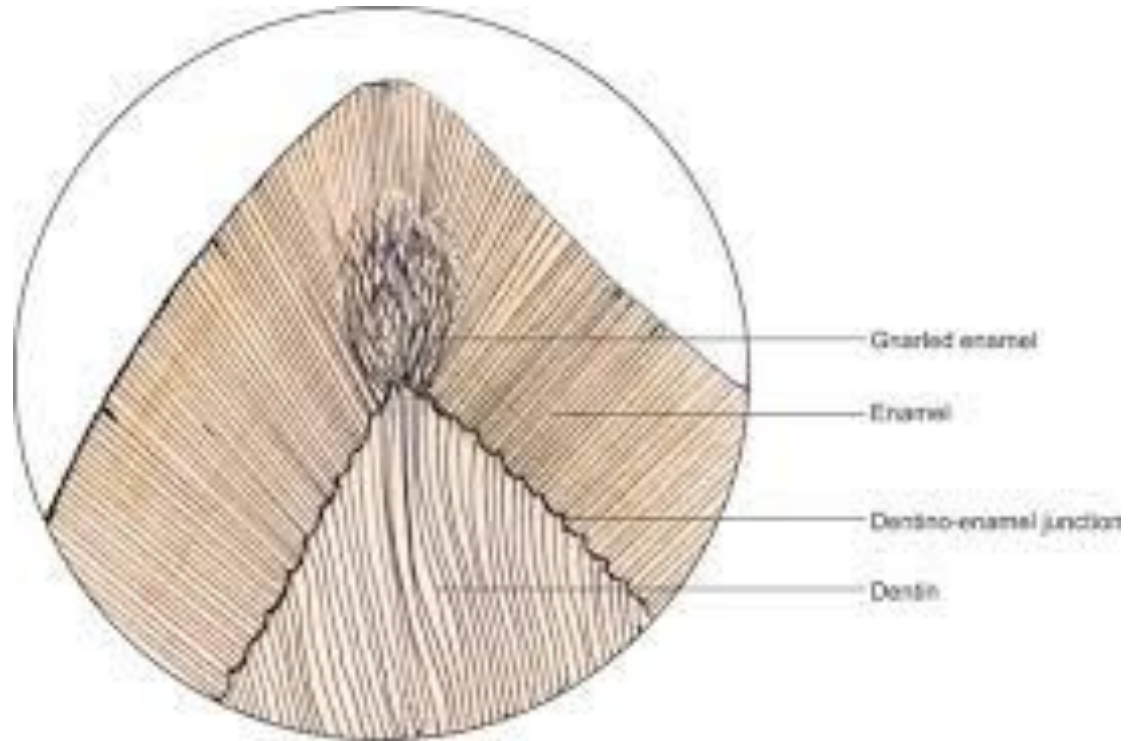
- Odontoblastic processes sometimes cross the dentinoenamel junction and get entrapped in the enamel matrix.
- They may serve as pain receptors, thereby explaining the enamel sensitivity experienced by some patients during cavity preparation.





GNARLED ENAMEL

The enamel rods at the cuspal and incisal region appear intertwined, twisted and inter twisted, and are more irregular. Such kind of optical appearance of enamel is called as gnarled enamel. They are more so at the cuspal region than incisal region.



ENAMEL PEARLS

Occasionally, the cells of the epithelial root sheath remain adherent to the dentin surface, they may differentiate into functioning ameloblasts and form small round islands of enamel. Such droplets of enamel are called enamel pearls. They may be found near or in the bifurcation or trifurcation of the roots of permanent molars.



PHYSICAL PROPERTIES

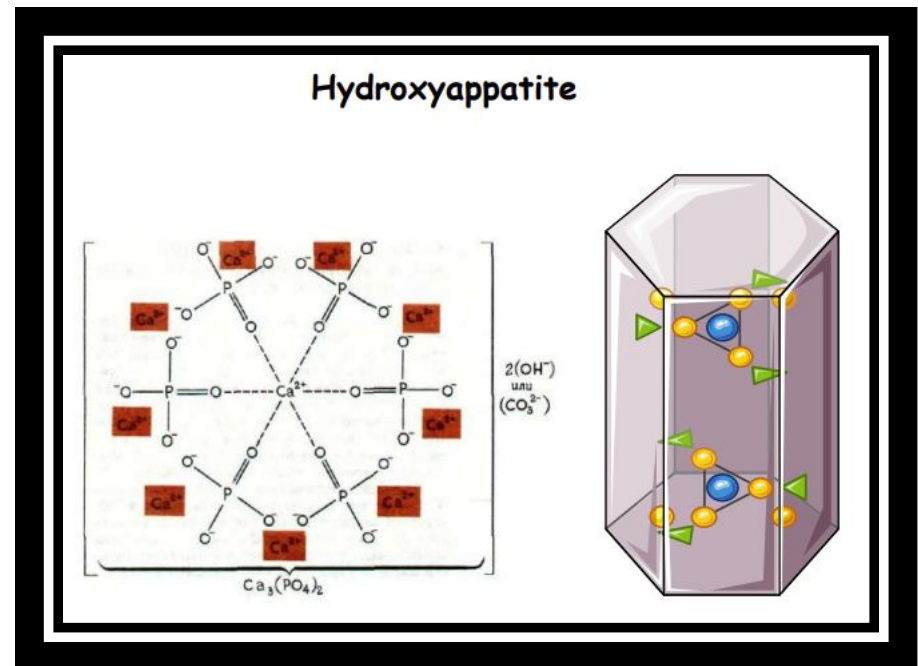
- Hard, brittle, totally acellular , highly mineralized
- Secretory product of stratified squamous epithelium
- Calcified tissue
- Hydroxyapatite crystal arrange in prism or rods
- Density: decreases from the surface of enamel to the dentinoenamel junction.
- Thickness: thickness over the cusps of the molars where it measures 2.5 mm & incisal edges of incisors where it is 2.0 mm

Chemical properties

- 96% inorganic - by weight
- 4% organic - by weight
- inorganic crystalline **calcium phosphate** – **hydroxyapatite**
- various ions like strontium, magnesium, lead and fluoride are present at some point during enamel formation
- In volume the organic matter and water are nearly equal to the inorganic contents.

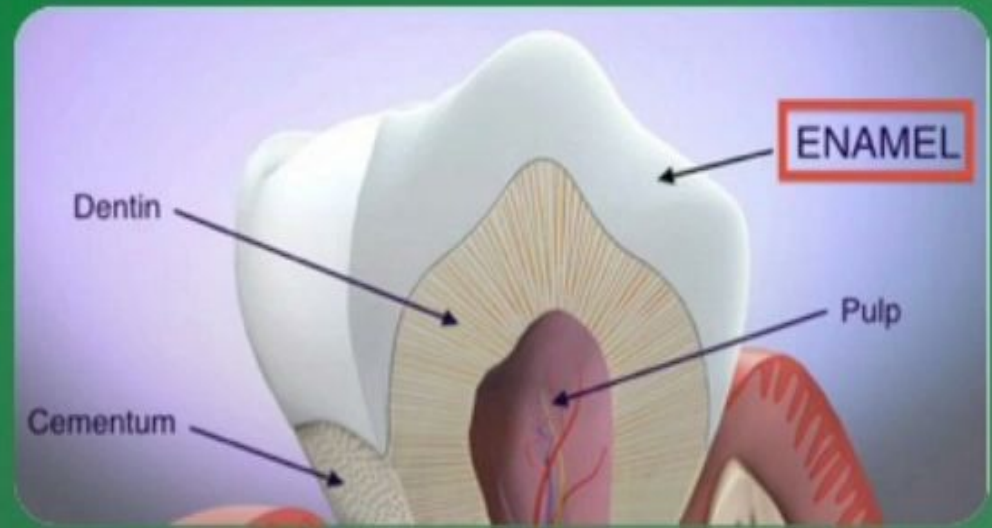
PROPERTIES	EXPLANATION
3. Color	From bluish white to yellowish or grayish white
4. Specific Gravity	2.8
5. Density	2.8 to 3 g/ml
6. Knoop Hardness Number	343
7. Hydroxyapatite crystal	300 times larger than dentin

- ENAMEL 343 KHN
- DENTINE 68KHN
- CEMENTUM 40KHN



- ENAMEL IS THE MOST HIGHLY MINERALIZED TISSUE OF THE BODY

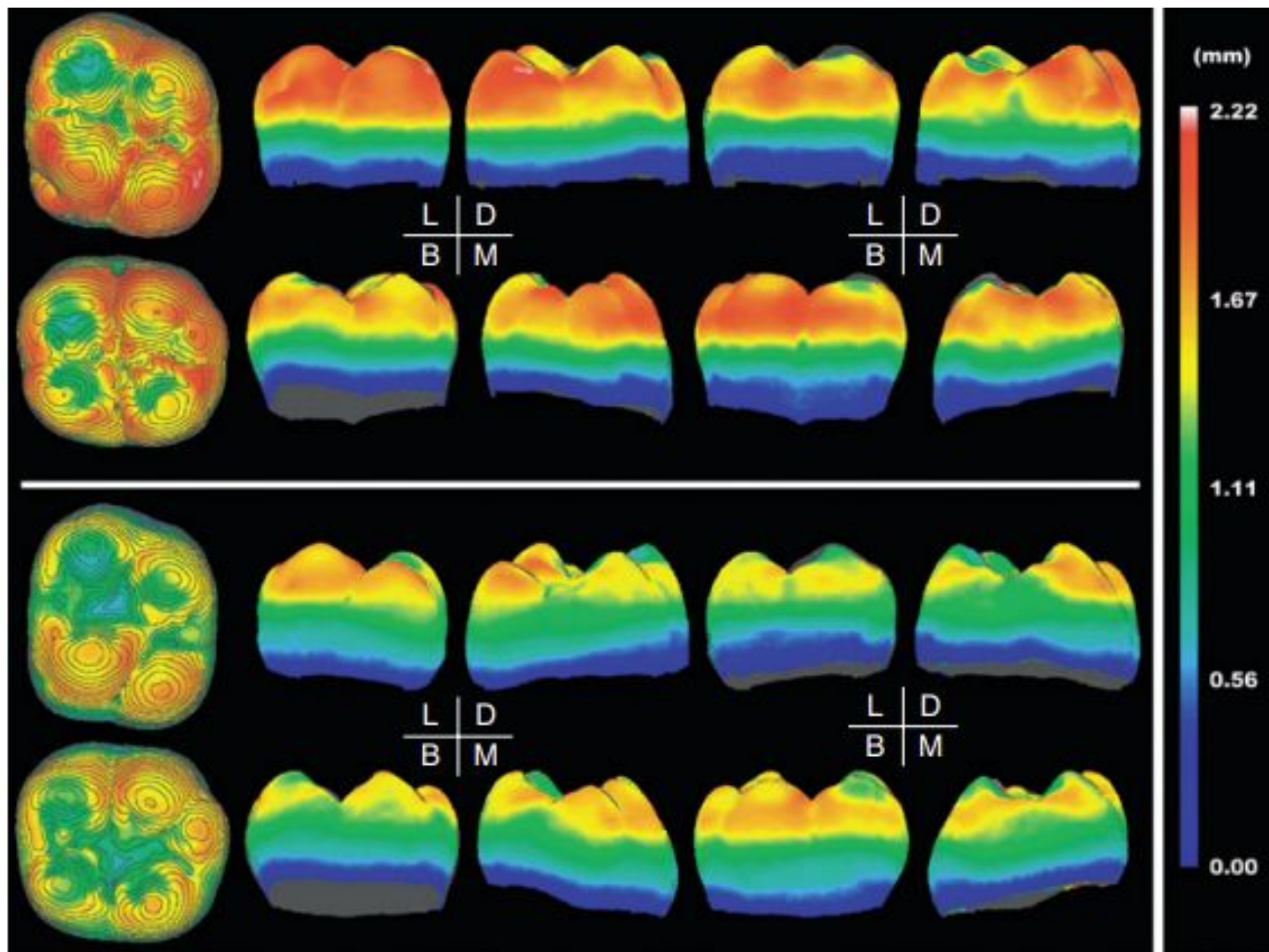
PROPERTIES	EXPLANATION
1. Permeability	Acts as Semi permeable
2. Thickness	2.5 mm at Cusp of Crown 100 micron at neck of tooth/Bottom of pit and fissure



Amelogenins (90%)

Non Amelogenins (10%)

(a) enamelin (b) ameloblastin and (c) tuftelin



Enamel

INORGANIC
MATERIAL

